

truth-preserving

not truth-preserving

Deduction

rules + premises
→ necessary consequence
truth-preserving

$R(A) \rightarrow B$

Induction

observations
→ general pattern / hypothesis
can be wrong

$A \dots A \rightarrow \text{pattern } S$

Abduction

observation + back-
ground knowledge
→ best explanation
can be wrong, useful

$OBS \rightarrow \text{best}(H)$

output is a hypothesis

Prediction

model + input
→ score / ranking
not entailment

$f(h,r,t) = 0.92$

output is a score